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Can We Prevent "Text Neck" Using Only a Smartphone? Real-Time Neck Angle Estimation and a Serious Game as a Case Study

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Abstract

Prolonged smartphone use often results in forward head posture, commonly known as "text neck," which places excessive strain on the cervical spine and may lead to various health problems. However, users typically find it difficult to monitor and correct their posture without external support. To address this challenge, we propose a method for real-time neck angle estimation using only a smartphone's front-facing camera and built-in sensors. Our approach extracts posture-related features such as the normalized distance from the nose to the neck base, interocular distance, smartphone tilt, and facial orientation. Regression analysis using Random Forest and Extra Trees models indicates that neck angle can be estimated with moderate accuracy (for example, $R^2 \approx 0.6$ and $MAE \approx 8.5^\circ$), although performance at extreme angles remains limited. To evaluate the practical utility of this system, we developed a serious game titled *Look Up and Tap!* that provides real-time visual feedback, including screen contrast adjustment and button highlighting, based on the estimated neck angle. A user study with 10 participants showed that this feedback significantly improved users' awareness of their posture and encouraged more upright neck positions. These findings suggest that neck angle estimation using only a smartphone is feasible and can help promote posture awareness in everyday settings. The serious game serves as one example of how this approach can be applied in real-world scenarios.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

posture correction, forward head posture, text neck, neck angle estimation, neck angle, biofeedback serious games, smartphone

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1 Introduction

A growing health concern known as "text neck" refers to a condition in which the natural curvature of the cervical spine becomes flattened. This condition is becoming increasingly common due to prolonged smartphone use. Under normal conditions, the cervical vertebrae form a gentle curve. However, with sustained forward head posture (FHP), they tend to align in a straight line, which increases mechanical stress on the neck. Such stress can lead to various physical and psychological problems, including neck stiffness, shoulder pain, appearance-related concerns, and discomfort.

FHP is typically observed when users bend their heads forward for extended periods to view objects below eye level, such as a smartphone screen. To investigate how common this behavior is, we conducted an online questionnaire in July 2023 through Yahoo! Crowdsourcing, targeting 2,000 Japanese participants. Yahoo! Crowdsourcing is a Japanese crowdsourcing service that has been confirmed to provide sufficiently reliable data [41]. Among the 1,623 valid responses, approximately 83% reported using their smartphones at positions below neck level. It is well established that greater neck flexion angles result in higher loads on the cervical spine [16], indicating that such usage habits can significantly contribute to the development of text neck.

To prevent this issue, users need to be aware of their posture while using smartphones. However, it is difficult to maintain continuous self-awareness in practice. A system that can estimate a user's neck angle in real time and provide immediate feedback may offer a practical solution.

Previous studies have proposed various methods for posture estimation. Some used wearable sensors attached to the back of the head [12], while others relied on external devices such as Microsoft Kinect or electric goniometers [1]. Although effective, these approaches require additional hardware, restrict usage environments, and reduce overall convenience. A notable prior attempt to estimate neck angle using only a smartphone was made by Lawanont et al. [25, 26]. Their method eliminated the need for external sensors, representing an important step toward practical posture monitoring. However, the system required the user's neck angle and face angle to match, which limited its applicability in real-world conditions for accurately estimating neck angle.

To provide a more practical solution that does not rely on external devices and functions under typical smartphone usage conditions, we propose a method that estimates neck angle using only a smartphone. This method imposes no strict constraints on the distance or orientation between the user's head and the device. It extracts posture-related features using the front-facing camera

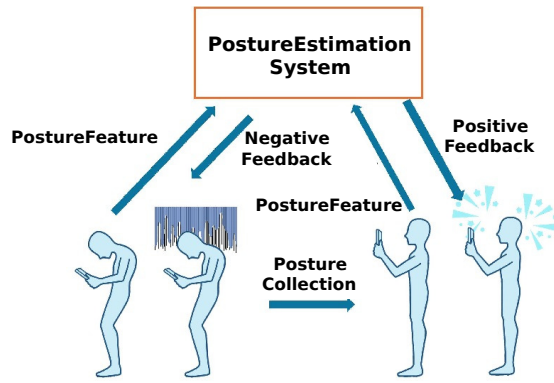


Figure 1: Conceptual overview of the proposed system. The smartphone estimates the user’s neck angle in real time and provides feedback, encouraging behavior change toward better posture.

and built-in sensors, including the ratio of the nose-to-neck distance to the interocular distance, smartphone tilt, and facial orientation. These features are then used to estimate the user’s neck angle. To evaluate the accuracy of this method, we also construct a dataset that spans a continuous range of smartphone tilt angles, facial orientations, neck angles, and face-to-device distances. This dataset is used to evaluate the accuracy of our estimation method.

Furthermore, to explore practical applications of neck angle estimation, we develop a serious game titled “Look Up and Tap!” The game provides real-time visual feedback based on the estimated posture, such as contrast adjustments and button highlighting, as shown in Figure 1. Players receive advantages when maintaining a good posture and face disadvantages when slouching.

Serious games are not only intended for entertainment but are also widely used for education, healthcare, and behavior change. Prior research has demonstrated their effectiveness in promoting healthy habits and improving user engagement [14, 17]. In this study, we utilize a serious game to examine whether real-time posture feedback can enhance users’ posture awareness and influence their engagement during smartphone use.

The contributions of this research are threefold:

- (1) We have proposed a neck angle estimation method that requires no additional hardware beyond a smartphone and is robust to variations in the user’s position relative to the device.
- (2) We have constructed a comprehensive dataset that systematically covers a wide range of neck, face, and smartphone orientations, as well as face-to-device distances.
- (3) We have demonstrated, through a serious game application, that real-time feedback based on neck angle estimation can significantly enhance users’ awareness of their posture.

2 Related Work

This section reviews prior research related to our study, focusing on four key areas: (1) the impact of smartphone use on neck strain, (2) posture correction systems developed for seated tasks such as computer work, (3) systems designed specifically for estimating and

correcting posture during smartphone use, and (4) the application of serious games for promoting health-related behavior change.

2.1 Smartphone Use and Neck Strain

Smartphone use is generally associated with increased neck angles. Lee et al. [30] reported that, in a laboratory study with 18 participants, they maintained head flexion angles of 33–45° (50th percentile angle) while using a smartphone. Furthermore, when performing three tasks (text messaging, web browsing, and video watching) in both standing and sitting postures, they found that text messaging was associated with the largest head flexion angles.

Numerous studies have explored the relationship between smartphone use and neck strain. Namwongsa et al. [36] conducted an experiment in which participants performed a 90-second text entry task at various neck angles while their neck muscle activity and perceived pain levels were recorded. The results indicated that greater neck flexion increased the activity of the cervical erector spinae muscles. Based on these findings, they suggested that a neck angle between 0° and 15° is optimal for smartphone use.

In a separate study, Namwongsa et al. [35] surveyed 799 Thai university students to examine the association between smartphone usage and musculoskeletal disorders. Over a 12-month period, they found that longer smartphone usage and smoking were significantly associated with cervical musculoskeletal disorders.

Kuo et al. [24] investigated postural differences across smartphone usage scenarios and found that head and neck flexion was greatest when users were seated without back support. They also demonstrated that a custom 3D-printed collar effectively reduced this flexion. Similarly, Lee [29] examined the relationship between FHP and various physical conditions such as range of motion, muscle endurance, pain, and disability. The study concluded that prolonged smartphone use in a forward head posture increases neck pain and interferes with daily life activities.

Building on these findings, our study aims to estimate users’ neck angles during smartphone use in order to help reduce the physical burden associated with prolonged forward head posture.

2.2 Posture Correction Systems for Seated Tasks

Various systems have been developed to support posture correction during seated tasks, such as computer work. Ishimatsu et al. [19] proposed BITAIKA, a system that detects forward-leaning posture using a side-mounted Kinect and identifies leg crossing via pressure sensors embedded in the seat. When poor posture is detected, the system provides feedback through pop-up windows. Experimental results showed a significant reduction in time spent in poor posture.

Park et al. [38] developed a system that uses a 3D camera to capture users’ side-view posture and provides EMG-based feedback, demonstrating its effectiveness. Similarly, Kim et al. [22] attached a 3D camera to a monitor to detect forward-leaning and found that visual feedback via pop-out windows helped prevent poor posture. Piñero-Fuentes et al. [39] used a webcam to estimate posture and applied a convolutional neural network (CNN) to classify it into four levels, from healthy to unhealthy, based on neck angle, shoulder alignment, and arm abduction. Choi et al. [10] proposed a movable monitor that gently tilts in response to the user’s posture, and their

50-minute experiment showed reduced forward head posture and fatigue compared to a fixed monitor.

While these systems are effective in fixed environments like desks, they are difficult to adapt to mobile contexts such as smartphone use. Our research addresses this gap by aiming to estimate posture in more dynamic and everyday settings.

2.3 Estimating Neck Angle During Smartphone Use

Many prior studies have investigated ways to estimate and correct neck posture during smartphone use. A common approach involves using external sensors to monitor neck angles. For instance, some systems attach accelerometers to the back of the neck [8], employ inertial sensor networks worn on the forehead, neck, and back [23], or combine magnetic sensors, accelerometers, and markers placed on the neck, ears, or shoulders [20]. Other approaches integrate sensors into wearable items such as neckband-style earphones [31], necklace-like devices [11], sensor-embedded garments [33], or cervical collars with embedded accelerometers and EMG sensors [27]. While these systems provide high accuracy, their dependence on specialized hardware limits their practicality and accessibility in daily life.

In contrast, several methods have been proposed that rely solely on smartphones. Gupta [15] used built-in accelerometers to infer posture, while Zindahi et al. [43] developed a warning system that activates when the phone's pitch exceeds 15° . However, because neck angle and phone tilt are not perfectly correlated, these methods may not provide sufficiently accurate estimations.

Lee et al. [28] introduced a system that combines smartphone tilt with features extracted from the front-facing camera, including face size, shape, and brightness. However, their approach assumes that the angle of the neck relative to the spine is the same as the angle of the face relative to the neck. This simplification can lead to inaccurate estimation of cervical strain. Arda et al. [4] proposed a monitoring system to prevent unhealthy posture when using a smartphone in the supine position. This system also utilizes the smartphone tilt angle and the front-facing camera to estimate the neck angle and the distance between the screen and the user's face. Similar to Lawanont et al. [25], it assumes that the neck angle and the face angle are identical, which limits the scenarios in which the neck angle can be accurately estimated.

Our work shares the same goal of estimating neck angle during smartphone use, but we aim to improve its practicality and generalizability. Specifically, we propose a method that uses only a smartphone without requiring any external devices, imposes no strict constraints on the spatial relationship between the user and the phone, and clearly separates smartphone tilt, head orientation, and neck flexion. By treating neck angle as the most relevant indicator of cervical strain and modeling it explicitly, our system provides a more accurate and flexible approach for posture estimation in everyday smartphone use.

2.4 Serious Games for Behavior Change

Serious games have been extensively used to support knowledge acquisition and promote behavioral change. For instance, Baños et al. [6] compared the effectiveness of the online game "ETIOBE Mates"

with traditional paper-based nutrition education for children. Their results showed that the game led to significantly greater improvements in nutritional knowledge. Similarly, Fuchslocher et al. [13] developed a game called "Balance" for teenagers with diabetes, finding that a version with an explicit health theme enhanced both enjoyment and self-efficacy compared to a version with implicit messaging.

Serious games have also been applied to posture improvement. Mihajlovic et al. [34] created a virtual reality game in which users catch butterflies to encourage neck movement, demonstrating that realistic, immersive environments help sustain user motivation more effectively than blank backgrounds. Beltran-Alacreu et al. [7] designed a rehabilitation game where older adults with chronic neck pain controlled a virtual airplane using head movements. The game was found to be both enjoyable and therapeutically beneficial. Roig-Maimó et al. [40] proposed a smartphone-based game in which the user manipulates a cursor through head movements to solve picture puzzles. They suggested that the game not only encourages engagement in rehabilitation and exercise but also allows for the automatic estimation of neck range of motion from the game interaction data. Youssef et al. [42] proposed a VR game designed to improve FHP based on the needs of both rehabilitation specialists and users. In a user study with three healthy adults, participants demonstrated high engagement with the exercises, although they did not necessarily find the experience enjoyable. The study also suggested that interventions using VR could be more motivating and appealing than traditional training methods. Asadzadeh et al. [5] similarly proposed a VR game designed to improve FHP. In a study involving 21 university students with FHP, the system achieved a high usability score on the System Usability Scale (SUS) of 87.14, and participants reported strong immersion and positive experiences on the Game Experience Questionnaire (GEQ). In addition, the sensors provided accurate measurements, and the system demonstrated high overall reliability, indicating that it can serve as an effective tool to support the correct and enjoyable practice of FHP correction exercises.

In a systematic scoping review, Andrew et al. [2] examined 21 studies that used serious games to promote health-related behaviors in youth. While some studies reported positive effects, others lacked rigorous validation or showed limited outcomes. The review emphasized the importance of thoughtful game design and user relevance in achieving meaningful behavioral change.

Building on this prior work, we develop a serious game that encourages better neck posture during smartphone use. Unlike previous systems that rely on VR or wearable sensors, our approach is highly accessible and runs entirely on a smartphone, making it suitable for everyday use.

3 Neck Angle Estimation Method

3.1 Overview

Our method estimates the user's neck angle using only a smartphone, relying on two key inputs: the tilt of the smartphone and the video feed from its front-facing camera. When interacting with a smartphone, the user's face is typically captured by the front camera, allowing the extraction of facial keypoints such as the nose, eyes, and the base of the neck.

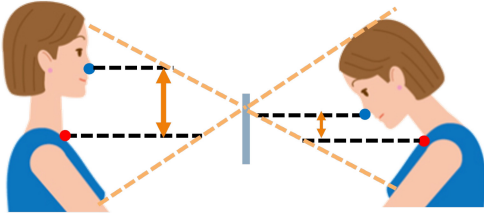


Figure 2: Illustration of the change in distance between the user's nose and the base of the neck as the neck angle changes.

Assuming that the smartphone remains stationary, the distance between the user's nose and the base of the neck in the captured image decreases as the neck tilts forward (see Figure 2). To account for individual differences in facial size and variations in face-to-camera distance, we normalize this measured distance using its value in a reference (upright) posture. The resulting ratio serves as a key feature for estimating neck angle.

3.2 Features and Preprocessing

As described above, the distance from the nose to the base of the neck provides useful information for estimating neck angle. However, this distance is also affected by several factors, including the orientation of the user's face relative to their neck, the position of the face relative to the smartphone, and the overall distance between the face and the phone. To ensure robust estimation, we extract and combine multiple features.

Specifically, we define the following features:

- $Angle_{face}$: Pitch angle of the user's face relative to the smartphone
- $Angle_{phone}$: Tilt angle of the smartphone relative to the ground
- $Dist_{eyes}$: Current distance between the user's eyes
- $Dist_{neck}$: Current distance from the user's nose to the base of the neck
- $Dist_{eyes}^{ref}$: Eye-to-eye distance in the reference (upright) posture
- $Dist_{neck}^{ref}$: Nose-to-neck distance in the reference (upright) posture

To adjust for changes in face-to-camera distance, we normalize the nose-to-neck distance using the following ratio:

$$Norm = \frac{Dist_{neck}}{Dist_{eyes}}, \quad Norm^{ref} = \frac{Dist_{neck}^{ref}}{Dist_{eyes}^{ref}}$$

This study focuses on estimating the neck angle in the sagittal (front-back) plane.

Accordingly, the final set of features used for estimation is:

$$F = \left\{ Angle_{face}, Angle_{phone}, \frac{Norm}{Norm^{ref}} \right\}$$

4 Dataset Construction

4.1 Limitations of Prior Dataset Construction Methods

Previous studies have often used subjective methods to define ground-truth neck angles. For example, participants were instructed to align their heads with wall-mounted markers. However, such methods fail to clearly distinguish between head and neck angles, raising concerns about the objectivity and accuracy of the resulting annotations.

Oyeleke et al.[37] labeled images as good (aligned head and neck) or bad (tilted forward or sideways) posture, and trained a CNN for binary classification. Lee et al.[28] constructed a dataset with four posture types by combining two body positions (seated and standing) with two head positions (upright and tilted), and used a smartphone to estimate neck angles. However, these studies lacked explicit criteria for what constitutes good posture, and their datasets were limited to coarse posture categories, making it difficult to capture the continuous variation in real-world smartphone use.

4.2 Our Dataset Construction Approach

To address these limitations, we designed a dataset that reflects the diversity of postures seen in everyday smartphone usage. Estimating neck angle in realistic conditions requires accounting for not only neck and smartphone orientation, but also the head-to-neck angle and the spatial relationship between the face and the phone. Exhaustively controlling all these variables would be impractical due to the large number of combinations and time required.

Our approach simplifies the collection process while preserving meaningful variation. We use an external camera to measure ground-truth neck angles and a smartphone to capture internal sensor data. These data streams are temporally aligned to ensure consistency across synchronized posture samples.

4.3 Ground-Truth Neck Angles and Reference Posture

There are two commonly used methods for defining neck angle based on anatomical landmarks. The first defines it as the inclination of the line connecting the seventh cervical vertebra (C7) and the tragus (the small cartilage protrusion in front of the ear canal). The second defines it as the inclination of the line connecting the acromion (the bony point where the clavicle meets the scapula) and the tragus. These anatomical landmarks are illustrated in Figure 3.

While both methods are widely used in posture assessment, the C7–tragus method often results in neck angle values of 10° – 20° even in relaxed, low-strain postures. Furthermore, such values tend to vary significantly between individuals. To ensure more consistent and interpretable labeling, we adopt the acromion–tragus definition, measuring the inclination of the line connecting these two points.

According to Ankrum et al. [3], the reference posture is defined as “the head/neck posture when looking at a visual target at eye level while standing upright,” in which the ear and shoulder are vertically aligned. Under our definition, this corresponds to a neck angle of 0° . In our study, we adopt this same reference, operationalized as the posture in which the acromion and tragus are vertically aligned

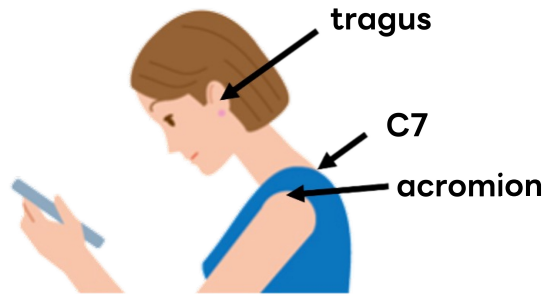


Figure 3: Anatomical landmarks used for neck angle estimation.

relative to the ground. To capture this reference posture during data collection, participants stood with their backs and heads against a wall.

4.4 Movement Design for Dataset Construction

To collect both feature values and corresponding ground-truth labels, dataset-construction participants were recorded from a lateral view while holding a smartphone and adopting a variety of postures. As discussed in Section 3, three primary factors influence the neck angle estimation: (1) the angle of the head relative to the neck, (2) the spatial relationship between the face and the smartphone, and (3) the distance between the face and the smartphone. Since all three are continuous variables, it was necessary to guide participants through a structured sequence of movements to ensure balanced coverage of the posture space.

To this end, participants followed a predefined set of movements, as illustrated in Figure 4. Each participant completed five movement sessions, each lasting approximately seven minutes and separated by short breaks, resulting in about 60 minutes of total recording time per person.

During the sessions, participants received real-time visual feedback on their current neck angle via a tablet display. Although exact angle matching was not required, participants were asked to approximate the target angles when prompted. Postures in which the smartphone was held above eye level were excluded from the dataset, as such positions typically exert minimal strain on the neck and are not the focus of this study.

4.5 System for Dataset Construction

We developed a data collection system that simultaneously captures both feature values and ground-truth neck angle measurements. The system comprises two primary components: one embedded in the smartphone for extracting feature data, and another using an external camera for capturing reference annotations.

The smartphone component records three types of feature data:

- (1) **Head orientation**, estimated from front-facing camera images using an end-to-end head pose estimation method [18]. Frames with a confidence score below 0.9 were excluded.

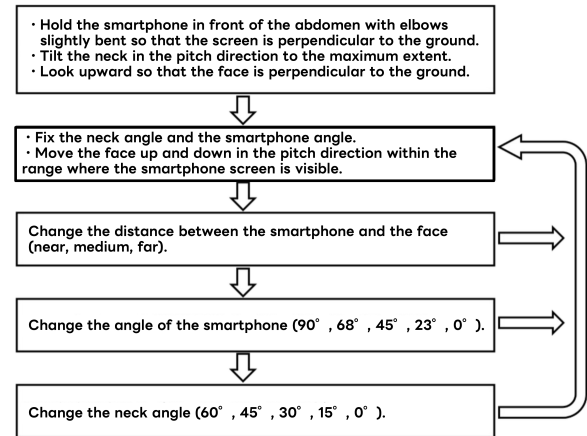


Figure 4: Movement sequence and instructions for dataset collection.

- (2) **Smartphone orientation**, measured via JavaScript-based access to built-in motion sensors.
- (3) **Skeletal keypoints**, extracted from the same front-facing camera images using PyTorch-OpenPose [9]. Frames were excluded if the confidence score was below 0.7 for eye or nose keypoints, or below 0.2 for the neck keypoint.

These data were sampled at a frame rate of 30 fps.

For the external camera component, a preliminary test using MediaPipe [32] indicated that shoulder keypoints were unstable. To obtain accurate neck angle annotations, we attached AR markers to the acromion and tragus. The external camera detected these markers and computed the neck angle as the inclination of the line connecting them. External recordings were captured at approximately 15 fps.

Since the two components operated at different frame rates, all data were synchronized using timestamps. Frame pairs were retained if their time difference was less than 40 milliseconds. When multiple matches existed, the pair with the smallest time gap was selected.

4.6 Dataset Collection Environment

Data collection took place in a well-lit indoor environment. To ensure consistent lighting conditions, blinds were closed, and partitions were installed to block visual interference from nearby individuals.

An Apple iPhone 11 was used to acquire smartphone-based data, while an IPEVO V4K USB document camera served as the external recording device. The external camera was positioned horizontally relative to the floor. The complete setup is shown in Figure 5.

4.7 Overview of the Constructed Dataset

Five participants (four male, one female), aged 20 to 22, were recruited from undergraduate and graduate programs. Individuals with visibly hunched shoulders were excluded, as such posture could misalign the anatomical markers (acromion and tragus), resulting in inaccurate ground-truth neck angle measurements. To

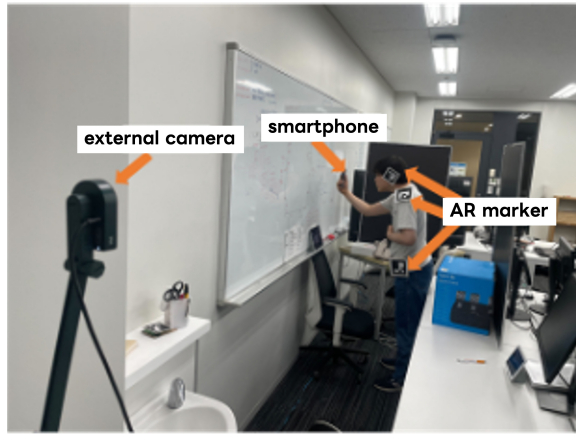


Figure 5: Data collection setup, including the smartphone, external camera, and AR markers.

minimize this risk, participants were instructed to avoid rolling their shoulders inward and to maintain an upright posture throughout the session.

Each participant contributed approximately 15,000 data points, yielding a total of 84,374 samples across all sessions.

5 Evaluation of Neck Angle Estimation Using Machine Learning

5.1 Estimation Model and Evaluation Setup

We trained a Random Forest regression model to estimate users' neck angles during smartphone use. This model was selected because it handles noisy input effectively, captures non-linear relationships, and resists overfitting. These characteristics are particularly important in our case, where features such as angles and keypoint distances may vary due to measurement error or user behavior.

Each participant completed five recording sessions. We used five-fold cross-validation, using four sessions for training and one for testing in each fold. This approach allowed us to evaluate the model's ability to generalize to new data from the same individual.

Before training, all features were standardized to account for differences in scale. We implemented the model using scikit-learn, a Python-based machine learning library.

5.2 Evaluation Results with Random Forest

The model achieved an average coefficient of determination (R^2) of 0.61, with a root mean square error ($RMSE$) of 11.64 and a mean absolute error (MAE) of 8.68. Table 1 presents the classification accuracy for neck angles grouped into seven categories, ranging from 0° to 60° in 10° increments. For most categories, the model correctly predicted the angle in approximately 40 percent of cases. Even when predictions were incorrect, they tended to be close to the actual angle. However, samples in the 0° category were frequently predicted as 10° .

We also performed per-participant evaluations using five-fold cross-validation. On average, the R^2 value was 0.60, with an $RMSE$

of 11.47 and an MAE of 8.42. As shown in Table 2, the most frequently predicted category usually matched the ground-truth label. Nevertheless, model accuracy varied considerably between individuals, as illustrated in Tables 3 and 4, with a difference of up to 15 percentage points. Interestingly, participants with lower overall accuracy often showed better precision in the 0° category.

5.3 Discussion of Random Forest Results

Both the group-level and individual-level predictions achieved R^2 values of approximately 0.6, indicating a moderate level of predictive accuracy. However, the root mean square error ($RMSE$) and mean absolute error (MAE) averaged around 8.5° , which is relatively high given the target range of neck angles from 0° to 60° . These results suggest that there is substantial room for improvement in the estimation accuracy.

When the ground-truth angles were grouped into categories, the model most often predicted the correct category or a neighboring one. This reflects a degree of consistency in the predictions. However, performance tended to decline at the extremes of the angle range. For instance, samples labeled as 0° were frequently misclassified as 10° . One possible explanation is that participants had difficulty maintaining a perfectly upright posture, especially at 0° . Slight inward rolling of the shoulders may have disrupted the alignment between anatomical landmarks, which would have influenced the accuracy of the ground-truth measurements. As described in Section 4.6, such shoulder misalignment can affect the true neck angle. Although we instructed participants not to roll their shoulders inward, we did not explicitly advise them to maintain a neutral or outward shoulder position. This lack of precise guidance may have introduced variation in posture.

Variations in model performance among participants could also result from differences in how skeletal features respond to changes in head posture. For some individuals, pose features may vary more clearly with changes in neck angle, while for others the change may be less pronounced. Other possible factors include facial structure, habitual posture, and inaccuracies in the reference posture, all of which may contribute to the variability in model accuracy.

Another limitation comes from the features used in the model. Our method focused only on pitch angles for both the head and the smartphone. However, roll and yaw were also present in practice and could affect measurements such as the distance between the nose and the neck or between the eyes. Adding roll and yaw to the feature set may improve the overall accuracy of neck angle estimation.

We also attempted to standardize the reference posture by asking participants to stand upright with their back and head against a wall while holding the smartphone in front of them. Even so, small tilts in the head or device were difficult to avoid completely. These slight deviations may have caused systematic errors, especially in distance-based features.

Finally, the accuracy of keypoint detection may have limited the model's performance. We used PyTorch-OpenPose to extract pose features, but its performance varied depending on the phone's orientation. As shown in Figure 6 and Figure 7, when the phone was held vertically, keypoints were usually detected accurately. In contrast, when the phone was nearly horizontal, the model often

Table 1: Classification probability for neck angle estimation using the Random Forest model (group-level evaluation).

ground-truth[°]	estimated angle[°]						
	0	10	20	30	40	50	60
0	0.16	0.58	0.18	0.04	0.02	0.01	0.00
10	0.05	0.46	0.33	0.10	0.04	0.01	0.00
20	0.02	0.24	0.41	0.19	0.11	0.03	0.01
30	0.00	0.07	0.20	0.30	0.32	0.10	0.01
40	0.00	0.03	0.11	0.18	0.37	0.28	0.04
50	0.00	0.02	0.05	0.08	0.25	0.41	0.18
60	0.00	0.01	0.01	0.03	0.13	0.39	0.43

Table 2: Average classification probability for neck angle estimation using the Random Forest model (per-participant evaluation).

ground-truth[°]	estimated angle[°]						
	0	10	20	30	40	50	60
0	0.03	0.70	0.24	0.03	0.00	0.00	0.00
10	0.01	0.44	0.42	0.12	0.01	0.00	0.00
20	0.00	0.19	0.47	0.24	0.08	0.00	0.00
30	0.00	0.05	0.21	0.36	0.33	0.06	0.00
40	0.00	0.02	0.10	0.20	0.40	0.27	0.01
50	0.00	0.01	0.05	0.09	0.28	0.46	0.12
60	0.00	0.00	0.01	0.03	0.14	0.46	0.36

Table 3: Classification probability for the participant with the highest accuracy using the Random Forest model.

ground-truth[°]	estimated angle[°]						
	0	10	20	30	40	50	60
0	0.23	0.59	0.17	0.01	0.00	0.00	0.00
10	0.09	0.55	0.31	0.04	0.01	0.00	0.00
20	0.05	0.24	0.51	0.14	0.05	0.01	0.00
30	0.00	0.03	0.12	0.59	0.19	0.06	0.01
40	0.00	0.01	0.05	0.18	0.36	0.36	0.04
50	0.00	0.00	0.01	0.07	0.18	0.52	0.22
60	0.00	0.00	0.00	0.01	0.01	0.35	0.63

Table 4: Classification probability for the participant with the lowest accuracy using the Random Forest model.

ground-truth[°]	estimated angle[°]						
	0	10	20	30	40	50	60
0	0.47	0.30	0.13	0.06	0.02	0.01	0.00
10	0.18	0.35	0.25	0.12	0.06	0.04	0.00
20	0.09	0.22	0.21	0.23	0.15	0.09	0.01
30	0.02	0.10	0.11	0.18	0.30	0.26	0.09
40	0.01	0.05	0.11	0.18	0.30	0.26	0.23
50	0.00	0.03	0.07	0.12	0.21	0.34	0.23
60	0.00	0.01	0.03	0.05	0.16	0.40	0.36

failed to detect the neck correctly and instead identified points near the upper chest. These detection errors likely reduced the overall accuracy of neck angle predictions.

5.4 Evaluation Using Extra Trees

To address the limitations observed in the Random Forest results, we examined whether a different model could provide better performance under the same experimental conditions. Because both the input features and ground-truth labels included noise and

potential outliers, we selected the Extra Trees model. This model is known for its robustness in handling data irregularities.

Applying the same training and testing procedure, the Extra Trees model achieved improved results. The average coefficient of determination (R^2) increased to 0.70. In addition, the root mean square error (RMSE) and mean absolute error (MAE) were reduced to 10.17 and 7.88, respectively. These metrics represent a moderate improvement compared to the Random Forest model. In terms of classification accuracy, the model most frequently predicted either the correct angle category or a neighboring one. Although accuracy

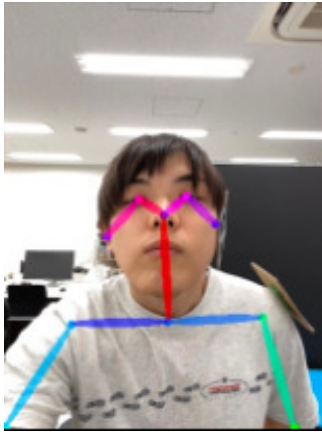


Figure 6: Example of successful skeletal keypoint detection by OpenPose, showing the accurate identification of base of the neck landmarks.

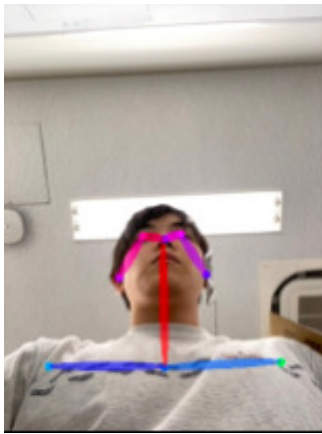


Figure 7: Example of unsuccessful skeletal keypoint detection by OpenPose, showing inaccurate identification of base of the neck landmarks.

remained slightly lower at the boundaries of the range, particularly at 0° and 60° , predictions for mid-range angles showed a slight enhancement.

When the model was trained and evaluated separately for each participant, performance improved further. The average R^2 rose to 0.75, while RMSE and MAE decreased to 9.03 and 6.92, respectively. The classification results followed a similar trend as those of the Random Forest model. This suggests that both models benefit similarly from personalized training using individual participant data.

5.5 Model Comparison and Discussion

The Extra Trees model demonstrated slightly better predictive performance than the Random Forest model, as shown by improvements in R^2 , root mean square error (RMSE), and mean absolute error (MAE). Despite these gains, the overall

accuracy remains insufficient for practical applications that demand high precision. To address this, additional techniques may be needed, such as application-specific threshold adjustments or the integration of multiple estimation approaches.

One notable limitation of the Extra Trees model was its reduced classification accuracy at the extreme neck angles, particularly at 0° and 60° . This outcome is likely related to the model's robustness to outliers, which makes it less likely to predict boundary values. Because both the most upright and the most slouched postures were predicted less frequently, neither model can be considered universally suitable. The optimal choice of algorithm should therefore be guided by the specific requirements and constraints of the intended application.

Given an average R^2 of approximately 0.70 and a mean absolute error around 7° to 8° , the current model may not be accurate enough for applications that require fine-grained posture estimation. Examples of such domains include clinical diagnostics or rehabilitation monitoring, where even small deviations can have significant implications.

However, the current level of accuracy may still be acceptable for scenarios where approximate posture information is sufficient. For example, systems that notify users when neck flexion exceeds a certain threshold for an extended period could remain effective despite occasional mispredictions. Similarly, applications that aim to provide general posture tracking throughout the day or promote user awareness through gamified feedback may benefit from this approach.

6 A Serious Game for Posture Correction: Look Up and Tap!

6.1 Feedback Design Based on Posture

To promote better posture during smartphone use, we developed a serious game in which the player's posture has a direct impact on gameplay. When the player maintains good posture, the game provides favorable visual effects. In contrast, poor posture introduces visual challenges that make gameplay more difficult. While feedback could also be delivered through sound effects or score bonuses, we chose visual feedback as the main channel for two key reasons: first, to offer continuous posture guidance during gameplay; and second, to minimize potential disturbances in public or shared environments where the game might be used.

Our feedback mechanism adjusts the color and contrast of game elements. When good posture is detected, the screen displays high-contrast visuals that are easy to recognize. On the other hand, slouched posture reduces the contrast, making it more difficult to distinguish important elements. This design encourages players to maintain an upright posture to improve gameplay performance.

In addition, when the user's posture falls within an acceptable range, the interface highlights key user interface elements such as the next button to be tapped. These visual cues provide further incentive for users to maintain appropriate posture.

One practical challenge is that smartphone users often hold the device close to their face, which can block the front-facing camera's view and interfere with posture detection. To address this issue, we implemented a rule that temporarily pauses the game if posture cannot be estimated for a certain period of time. This

Table 5: Classification probability for neck angle estimation using the Extra Trees model (group-level evaluation).

		estimated angle[°]						
		0	10	20	30	40	50	60
ground-truth[°]	0	0.03	0.70	0.24	0.03	0.00	0.00	0.00
	10	0.01	0.44	0.42	0.12	0.01	0.00	0.00
	20	0.00	0.19	0.47	0.24	0.08	0.00	0.00
	30	0.00	0.05	0.21	0.36	0.33	0.06	0.00
	40	0.00	0.02	0.10	0.20	0.40	0.27	0.01
	50	0.00	0.01	0.05	0.09	0.28	0.46	0.12
	60	0.00	0.00	0.01	0.03	0.14	0.46	0.36

encourages players to avoid using postures that prevent accurate tracking and reinforces the intended behavioral goal of maintaining good posture.

6.2 Game Mechanics

To enhance the effectiveness of the feedback strategy described in Section 6.1, we developed a game that requires continuous visual attention and precise motor control. The core activity is a “Number Tap” task, in which players tap numbered buttons in ascending order as quickly as possible. This task requires visual scanning and cognitive focus, and has been linked to improvements in speed reading and reaction time in athletic contexts [21].

In our implementation, the numbered buttons function as dynamic targets that respond visually to the player’s posture. When posture worsens, the screen contrast decreases, making it harder to identify the correct number. In contrast, maintaining good posture results in the next correct number being visually emphasized, which simplifies the task and positively reinforces proper posture.

To ensure game fairness and discourage random tapping, we introduced the following mechanics. When the correct number is tapped, the corresponding button changes color to indicate success. If the player taps an incorrect number, a two-second penalty is applied during which all buttons are temporarily disabled. This mechanism promotes intentional and posture-aware interaction while discouraging careless input.

6.3 Calibration

Accurate estimation of head posture requires a reference pose that reflects the user’s ideal alignment. Before starting the game, users are instructed to assume a proper posture on a calibration screen, as shown in Figure 9. To minimize the influence of smartphone tilt and facial orientation variability, the system captures this reference pose only when certain conditions are met.

In this study, the reference conditions are defined as follows:

- (1) The smartphone’s orientation, measured using its three-axis gyroscope (alpha, beta, gamma), must be nearly perpendicular to the ground and directly facing the user. The specific criteria are:
 - $-3 < ((\alpha + 180) \bmod 360) - 180 < 3$
 - $-3 < \beta - 90 < 3$
 - $-3 < \gamma < 3$
- (2) The user’s facial orientation relative to the device (pitch, roll, yaw) must be within plus or minus 3° .

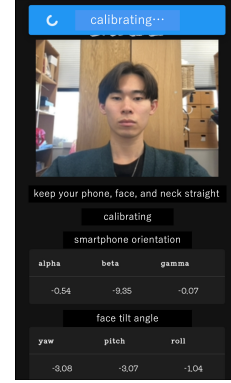


Figure 8: Calibration screen for establishing a reference posture.

- (3) The system must successfully detect the skeletal structure from the eyes to the neck.

6.4 System Architecture

To implement the gameplay and posture-feedback system described above, we developed a browser-based web application. As shown in Figure 10, the system consists of three main components: (1) a frontend user interface that handles gameplay interactions, (2) a game API that manages user information and scores, and (3) a posture API that estimates head angles using sensor readings and camera images.

The game API is implemented as a RESTful service supported by a lightweight database. In contrast, the posture API performs more computationally intensive operations, such as image processing and machine learning inference. To ensure smooth execution, this component is designed to run on infrastructure equipped with a GPU. The code for the posture API is available on GitHub¹, where details such as the hyper-parameters of the posture estimation model can also be found.

As discussed in Section 5, the Extra Trees model achieved slightly better overall accuracy than the Random Forest model. While Random Forest performed more accurately at certain boundary angles like 0° and 60° , we determined that this issue could be addressed by fine-tuning the threshold used for visual feedback during gameplay. This approach helps minimize the impact of

¹<https://github.com/kntWT/posture-correction-backend/tree/ozchi-2025>

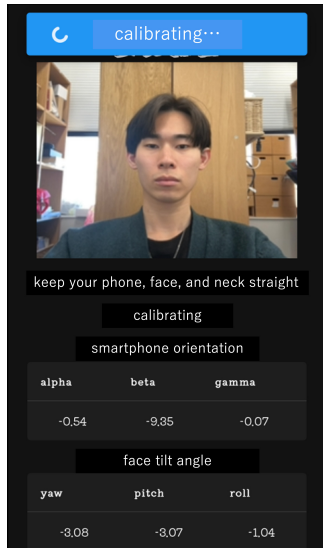


Figure 9: Calibration screen for establishing a reference posture.

prediction errors on user experience. Based on these considerations, we adopted the Extra Trees model for neck angle estimation in our system.

7 User Study: Evaluating Posture Feedback through Gameplay

7.1 Overview

To examine the impact of real-time posture-based feedback on users' awareness and overall experience during gameplay, we conducted a user study guided by the following hypotheses:

- **H1:** Providing feedback that makes the game easier when the user's posture is good and more difficult when it is poor will increase the user's awareness of maintaining proper posture.
- **H2:** Game difficulty that adjusts interactively based on the user's posture, becoming easier with good posture and more difficult with poor posture, will enhance the user's enjoyment of the game.

The study compared two conditions: one with feedback based on the user's neck angle and another without such feedback. After each session, participants rated their enjoyment and their awareness of posture.

We defined the two experimental conditions as follows:

- **Face Recognition Feedback (FaceFB):** Feedback was provided only when the system failed to detect facial features such as the eyes or the base of the neck. In these cases, an alert was shown indicating a face recognition failure.
- **Posture Recognition Feedback (PostureFB):** Feedback was provided in real time based on the estimated neck angle. Additionally, alerts were shown when posture estimation failed, in the same manner as in the FaceFB condition.

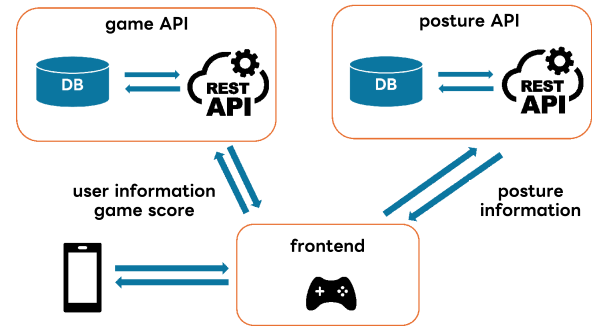


Figure 10: System architecture of the smartphone-based neck angle estimation and serious game application.

Even when face recognition succeeded, posture estimation was considered to have failed if the neck region was not clearly visible or if detection confidence was low.

To control for individual differences in gaming familiarity and posture habits, we employed a within-subjects design. Each participant completed five trial blocks, with each block consisting of gameplay under the FaceFB condition followed by the PostureFB condition, and then a subjective evaluation.

Before starting the trials, participants completed a practice session consisting of two rounds under each condition, without any evaluations. This allowed them to become familiar with the game mechanics and feedback system.

Ten participants (nine male and one female), all aged between 19 and 23, were recruited from undergraduate and graduate programs. All participants reported using smartphones regularly. Three of the male participants had also taken part in the dataset construction phase.

In this study, the contrast between the button text and background varied across six levels depending on the estimated neck angle, with thresholds set at 10° intervals from 0° to 60°. This design was informed by the accuracy characteristics of the estimation models discussed in Sections 5.4 and 5.5. The goal was to tolerate some prediction errors while still offering responsive feedback that could raise users' awareness of their posture.

Additionally, the threshold for highlighting the next button was set at 15°. This was based on previous findings [36] indicating that neck angles between 0° and 15° are optimal for minimizing strain during smartphone use. We judged that the estimation model could reliably approximate angles near this range, making it a suitable threshold for posture-based feedback.

7.2 Procedure

The procedure of this user study is illustrated in Figure 11. After receiving an explanation of the study, each participant underwent a calibration process as described in Section 6.3. Once the reference posture was recorded, the system displayed the estimated neck angle at the center of the screen (see Figure 9), allowing participants to observe how their posture influenced the angle estimates.

Participants then completed two practice rounds of gameplay, one under the FaceFB condition and one under the PostureFB

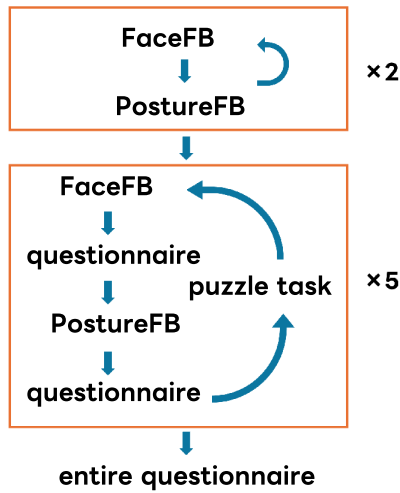


Figure 11: Flow of the user study procedure.

condition, and repeated this sequence once. No subjective ratings were collected during the practice phase to avoid learning bias.

The main session consisted of five sections. In each section, participants played one round under the FaceFB condition and provided subjective ratings, followed by one round under the PostureFB condition and a second set of ratings. The order of conditions was fixed, with the FaceFB condition always preceding the PostureFB condition. This ordering was chosen to avoid carryover effects: if participants experienced posture-based feedback first, they might remain posture-aware even in subsequent no-feedback trials, potentially affecting the results.

Figure 12 presents example screens from each feedback condition. It includes visual states corresponding to good, neutral, and poor posture in the PostureFB condition, as well as the feedback interface in the FaceFB condition and the screen shown when posture estimation failed. In the PostureFB condition, the next target button was highlighted only when the estimated neck angle was within 15° (see Figure 12a).

All gameplay was conducted with participants standing and holding the smartphone. To reduce residual effects between sections, participants completed a three-minute seated block puzzle task between sections, which required spatial rearrangement of pieces within a fixed frame.

After completing all five sections, participants answered a final questionnaire. The entire session lasted approximately 45 minutes. The system's end-to-end latency, measured from the moment the face image was captured to the moment the estimated neck angle was returned, ranged from approximately 0.5 to 1.0 seconds using an NVIDIA GeForce RTX 4090 GPU.

7.3 Results

Figures 13 and 14 show participants' self-reported ratings of posture awareness and game enjoyment for each condition and session, using a five-point Likert scale. The results indicate that posture awareness was higher under the PostureFB condition, while game enjoyment did not differ significantly between the two conditions.

A paired t-test was conducted on the ratings from each section under the FaceFB and PostureFB conditions ($n = 5 \times 10 = 50$). The analysis revealed a statistically significant increase in posture awareness in the PostureFB condition ($p < .001$).

To further analyze how users' posture was estimated, we evaluated the estimated neck angles across sessions. To assess overall posture differences while excluding extreme values, we focused on the central 80% of the data by trimming the top and bottom 10% of neck angle values. This decision was based on the distribution of estimated angles. Histograms showed isolated outliers forming minor peaks, suggesting that a small number of frames included anomalous values, likely caused by transient recognition errors or momentary occlusions. By removing these extremes, we aimed to obtain a more representative measure of typical posture. Figure 15 shows box plots for each condition based on these trimmed means. A paired t-test showed that participants exhibited significantly better posture (i.e., smaller neck angles) under the PostureFB condition ($p < .001$).

In the PostureFB condition, feedback was also provided by changing the text color of the next button when the neck angle was estimated to be within 15° , with the goal of encouraging proper posture. Figure 16 shows the proportion of time during which the neck angle remained within 15° under each condition. The proportion was significantly higher in the PostureFB condition ($p < .001$), indicating that the feedback successfully promoted better posture maintenance.

In the final questionnaire, participants described the posture-based feedback as novel and engaging. However, many also reported experiencing fatigue, particularly from maintaining posture and holding the device at eye level. Most participants preferred text color changes over background color modulation for feedback. Some noted that the clear highlighting of the next button made the game feel too easy, which in turn reduced their enjoyment.

7.4 Discussion of Posture Awareness

The results presented in Section 7.3 support H1. Feedback based on neck angle estimation effectively increased participants' awareness of their posture. This effect is likely due to the combination of real-time visual feedback and the motivational structure of the game, in which maintaining good posture led to improved playability.

In addition, the estimated posture data showed that participants maintained significantly better posture in the PostureFB condition. This suggests that increased awareness resulted in measurable behavioral changes. The higher proportion of time during which neck angles remained within 15° further supports the idea that feedback guided users to maintain posture within a specific range. Dynamically adjusting this threshold could enhance posture control in future implementations.

Participants generally reported that posture estimation felt accurate. However, some expressed confusion about why they were sometimes rated as having poor posture or noted that small neck movements caused sudden changes in feedback. These concerns point to issues with sensitivity or stability in the system, which may affect user trust and overall usability.

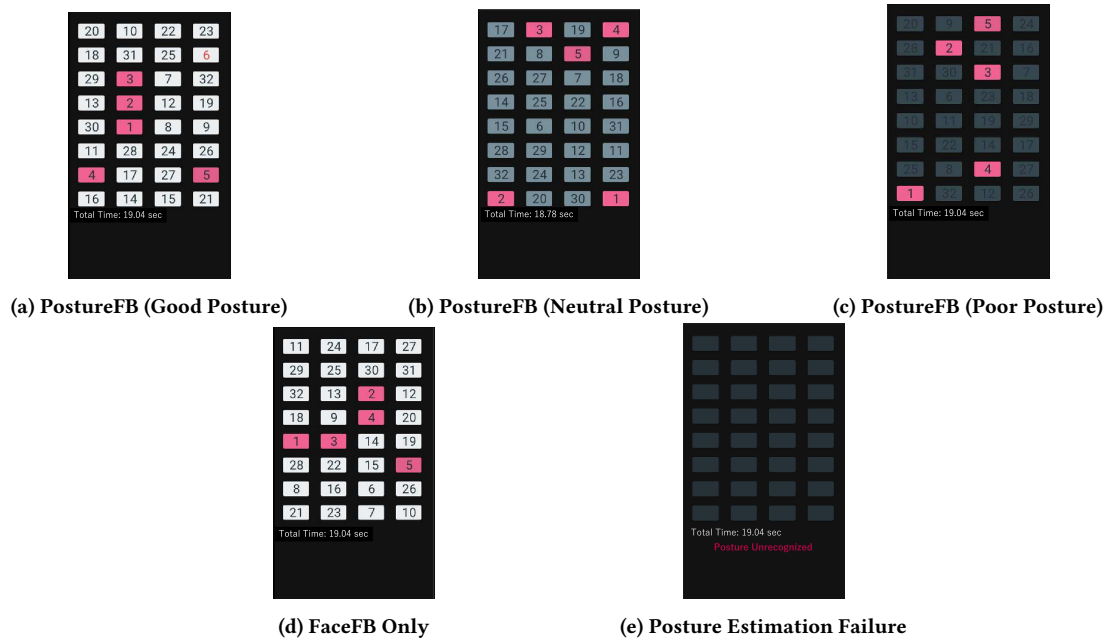


Figure 12: Gameplay screens

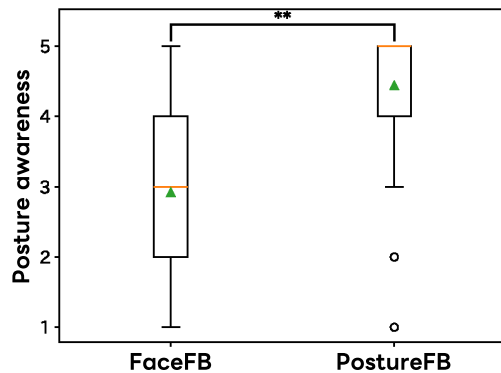


Figure 13: Participants' posture awareness ratings for FaceFB and PostureFB conditions.

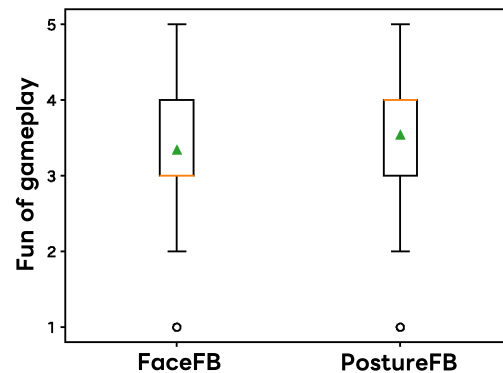


Figure 14: Participants' game enjoyment ratings for FaceFB and PostureFB conditions.

Interestingly, some participants remained posture-aware even during the FaceFB condition, which lacked posture-specific feedback. This may reflect a carryover effect from earlier exposure to the PostureFB condition. As shown in Figure 17, posture awareness remained high across sessions in the PostureFB condition and gradually increased in the FaceFB condition. These patterns suggest that feedback-induced awareness may persist beyond the feedback itself, indicating some durability in the effect.

Figure 18 shows the average neck angle (excluding extreme values) across participants. In the FaceFB condition, neck angles gradually decreased over time, while in the PostureFB condition they remained relatively stable. This suggests that although

participants retained awareness, they may have found it difficult to sustain proper posture without real-time support, possibly due to fatigue or lack of guidance.

Several participants mentioned physical strain, with comments such as “maintaining proper posture was tiring” or “holding the smartphone at eye level made my arms tired.” These responses indicate that the experiment protocol may need adjustment, such as reducing the number of trials or adding longer breaks, to better isolate posture awareness from fatigue effects, and also suggest that such arm fatigue could make it difficult to maintain posture for extended periods.

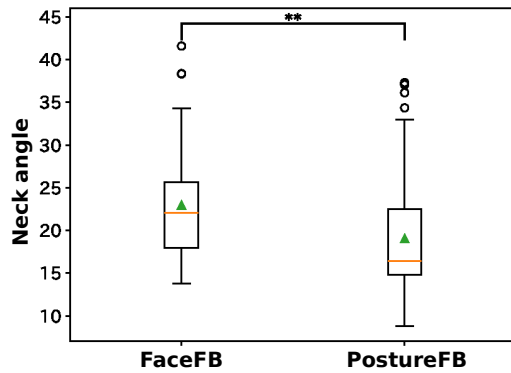


Figure 15: Average estimated neck angle (central 80%) for FaceFB and PostureFB conditions across participants.

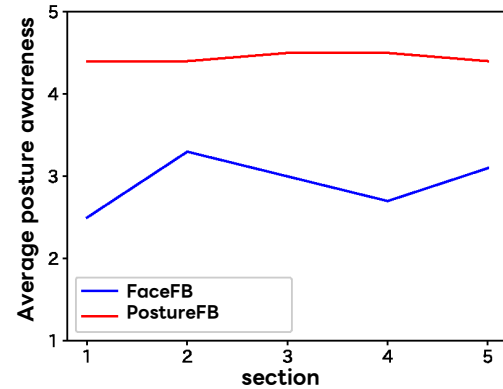


Figure 17: Average posture awareness across sessions for FaceFB and PostureFB conditions.

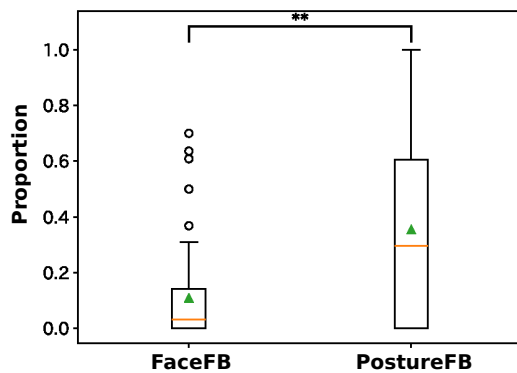


Figure 16: Proportion of frames with neck angles within 15° for FaceFB and PostureFB conditions.

Additionally, even in the FaceFB condition, gameplay was paused if the system could not detect the user’s face. This may have encouraged participants to keep their face and neck in view, indirectly promoting posture awareness. Future work may consider alternative sensing strategies, such as side-view tracking, to better assess users’ natural posture without requiring direct camera alignment.

7.5 Discussion of Game Enjoyment

Contrary to Hypothesis 2, providing feedback based on neck angle did not significantly enhance participants’ enjoyment of the game, as reported in Section 7.3. One plausible explanation is that the game itself was too simple, leading to a sense of monotony over repeated sessions. This interpretation is supported by participants’ comments such as “the gameplay was too simple; it would be more engaging with some additional features”, “a more complex puzzle might be more enjoyable”, and “the game became boring quickly, although the time attack element did make me want to improve my

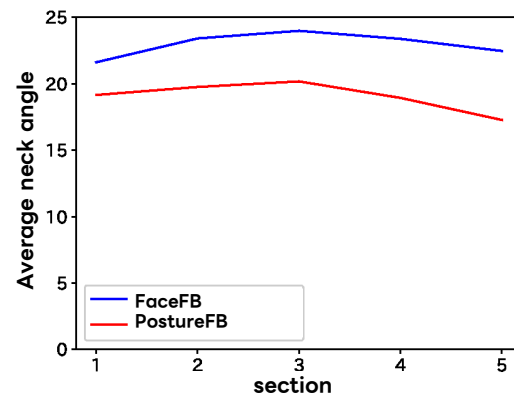


Figure 18: Average neck angle across sessions for FaceFB and PostureFB conditions.

score”. These responses suggest that enriching the game content could lead to greater enjoyment.

Another factor that may have influenced enjoyment is the accuracy of posture estimation and the total time required to complete each game session. Figure 19 shows a scatter plot of enjoyment scores and completion times, indicating a moderate negative correlation ($r = -0.52$). This suggests that longer completion times were generally associated with lower enjoyment.

Participants also expressed frustration with recognition failures. Comments included, “when my face was not detected for an extended period, it became frustrating, as I spent more than half of the game unable to interact”, and “perhaps it was my fault, but I was often not recognized. I would enjoy the game more if it worked more reliably”. These remarks highlight how system reliability directly affected the overall user experience.

In summary, enjoyment appears to have been shaped by multiple factors, including the simplicity of the game, the robustness of

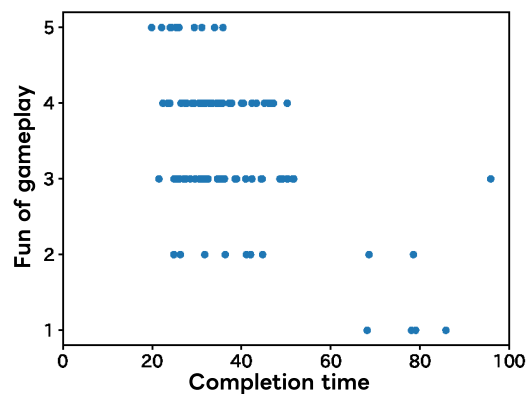


Figure 19: Scatter plot showing the relationship between game enjoyment scores and completion times.

posture estimation, and the duration of each task. These factors may have overshadowed any potential benefits of posture-based feedback. Future studies may improve outcomes by increasing gameplay complexity, enhancing system stability, and refining the experimental procedure to better isolate the specific impact of feedback on enjoyment.

8 Conclusion

Although poor posture during smartphone use is a common issue, most existing monitoring systems either rely on external hardware or lack sufficient accuracy in estimating neck angles, which limits their practicality. In this study, we proposed a method for estimating neck angle using only a smartphone by leveraging its front-facing camera and built-in sensors. Our approach focuses on the distance between the user's nose and the base of the neck, extracted from camera images. To address limitations in previous data collection methods, we also developed a streamlined dataset construction process that maintains sample size while reducing procedural complexity.

Using the constructed dataset, we performed multiple regression analysis with a Random Forest model. The results showed a coefficient of determination around 0.6 and a mean prediction error of approximately 8.5° , both when training on data from all participants and when training individually. Furthermore, when classifying neck angles into six categories at 10° intervals from 0° to 60° , the model most often predicted the correct class, and misclassifications typically occurred between adjacent categories. To address accuracy limitations caused by shoulder movement and feature instability, we applied the Extra Trees model, which improved overall accuracy but reduced precision at extreme angles. These results suggest a need for model selection and feedback design tailored to the target application.

To explore the behavioral potential of real-time posture estimation, we developed a serious game that visually responds to neck posture. Our user study showed that posture-linked feedback increased participants' posture awareness and encouraged the maintenance of appropriate neck angles. However, the feedback

did not significantly improve enjoyment, possibly due to gameplay simplicity and recognition errors.

We believe that posture-based feedback systems like ours can be extended to everyday smartphone use cases such as casual gaming or social media, helping users build sustainable awareness of their posture in natural settings.

Limitations and Future Directions. This study has several limitations. First, the dataset was collected under controlled indoor conditions with a small and demographically narrow sample, particularly biased in terms of age and gender, which may limit generalizability. Second, although the Extra Trees model improved robustness, it still showed reduced accuracy at extreme neck angles. Third, the game was intentionally simplified to isolate the effect of feedback, which may have diminished its entertainment value. Lastly, the feedback relied solely on visual cues, which may not suit all users or contexts.

To address these limitations, future work should expand data collection to naturalistic environments with more diverse participants, explore multi-modal feedback (e.g., haptic, auditory), and improve system robustness. Additionally, incorporating real-time user modeling could enhance estimation accuracy and enable more personalized and engaging feedback strategies. Beyond serious games, we also envision applying this feedback mechanism to everyday smartphone use cases, such as casual gaming or social media browsing, to support sustainable posture habits in natural settings.

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